

DTS 201 · LECTURE 03

Data Collection, Sampling & Quality

Learning Objectives

- Distinguish primary from secondary data, and pick a collection method to match a question
- Explain the difference between a population, a sampling frame and a sample
- Apply four probability techniques: random, systematic, stratified, cluster
- Recognise sampling bias — and name the five ways it creeps in
- Audit a dataset on six quality dimensions using `pandas`

01

Data Collection

Collection is a design decision

Data collection is the deliberate process of gathering observations so that a specific question can be answered.

Note the word **deliberate** . Who you ask, when you ask, and what you choose to record are all decisions — and every one of them is inherited by the analysis.

No model, however clever, repairs data that was gathered from the wrong people.

FOUR QUESTIONS, BEFORE YOU GATHER ANYTHING

- 01 What exactly am I trying to find out?
- 02 Who or what does that question apply to?
- 03 Which observations would actually settle it?
- 04 Am I allowed to record them, and does the person know?

Different data collection sources

01

Surveys

Fixed questions, many people.
Cheap to scale, weak on the
“why”.

02

Interviews

Deep, open, few people. Rich
reasons; slow and hard to
compare.

03

Observation

Watch what people do rather
than what they say they do.

04

Experiments

Change one thing on purpose,
hold the rest still. The only
route to cause.

05

Sensors & devices

Automatic, continuous, high
volume. Silent when the device
fails.

06

Web scraping

Pull public pages into a table.
Fragile, and check the terms of
use.

07

APIs

A structured feed the owner
intends you to use. Usually
JSON.

08

Administrative records

Registrations, transactions,
attendance. Collected for
admin, reused for analysis.

Data Collection Methods

Primary: You collected it yourself

Gathered first-hand for your own question, by you or your team.

Examples: a survey you ran, lab experiment, interviews, etc.

Strength: fits your question exactly; you know every decision made

Cost: slow, expensive, needs ethical clearance

Secondary: Someone else collected it

Already gathered, for a purpose that was probably not yours.

Examples: national statistics, Kaggle datasets, company records, open APIs

Strength: fast, cheap, often very large

Cost: you inherit their definitions, their gaps and their bias

When using secondary data always ask this: who collected the data, from whom, when, and what were they hoping to find?

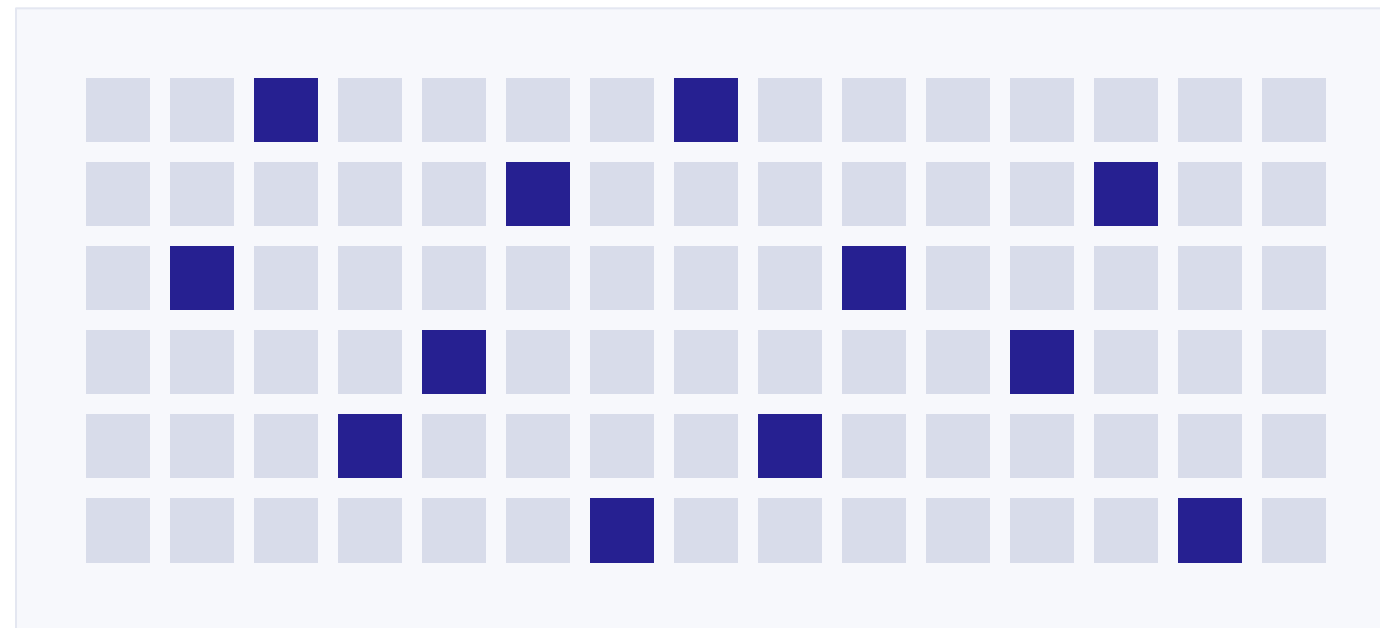
Trade-offs of different data collection sources

SOURCES	COST	SPEED	CONTROL	BIGGEST RISK
Survey	Low	Fast	Medium	Only motivated people reply
Interview	High	Slow	High	The interviewer is prone to guide the answer
Experiment	High	Slow	Highest	Results from lab experiments may not fit into the real world.
Sensors	Medium	Instant	Low	Drift and silent outages
API / open data	Very low	Instant	None	Definitions can change without warning
Scraping	Low	Medium	Low	Could break, may have limits or be fully prohibited.

02

Data Sampling

Population and sample



■ population has 90 units ■ sample is 12 units

A **population** is every unit your question is about. A **sample** is the subset you actually measure.

We almost never measure the whole population. It is too large, too expensive, or partly unreachable.

So the entire game is this: choose a sample small enough to afford, and honest enough that what is true of it is true of the population.

Why not just measure everyone?

Cost

A full census of a country runs into hundreds of millions. A national sample survey of ~2,000 people costs a rounding error of that.

Time

Censuses are run once a decade. A sample can be fielded, cleaned and reported inside two weeks.

Reach

Some populations cannot be fully enumerated at all — every fish in a lake, every future customer.

Care

Fewer units means more attention per unit — better questions, better checks, fewer recording errors.

THE PAYOFF A small, well-drawn sample beats a huge, carelessly drawn one. Size is not what makes a sample honest — method is.

Key terms

Population	Every unit the question is about.	E.g all 8,400 students enrolled a session.
Sampling frame	The practical list you can use to select your study participants.	Of the 8,400 only 7,900 have their email addresses in the school DB.
Sample	The participants you actually measure from the sampling frame	We decide to contact 400 students out of the 7,900 and collect their CGPAs.
Parameter	A number describing the <i>population</i> . Usually unknown.	Say mean (μ): The true mean CGPA of all 8,400, is unknown because we don't have the data.
Statistic	A number describing the <i>sample</i> . Your estimate of the parameter.	Mean CGPA of the 400, written $\bar{x}$ is known.

Sampling techniques - Probability and Non-Probability

PROBABILITY

Chance decides who is in

Every unit has a known, non-zero probability of being selected. Selection is left to a random mechanism, not to a person.

Approach:

simple random

stratified

systematic

cluster

NON-PROBABILITY

A human decides who is in

Selection depends on availability or judgement. Some units have zero chance at all of being chosen.

Approach:

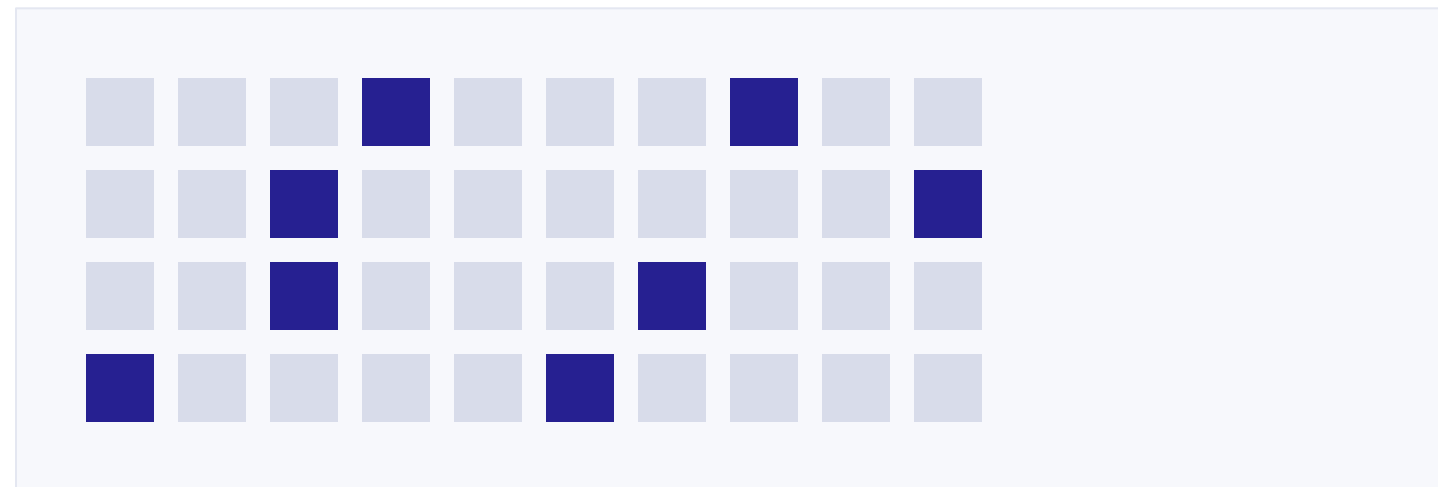
convenience

purposive

quota

snowball

TECHNIQUE 01 Simple random sampling



No pattern, no grouping just pure chance

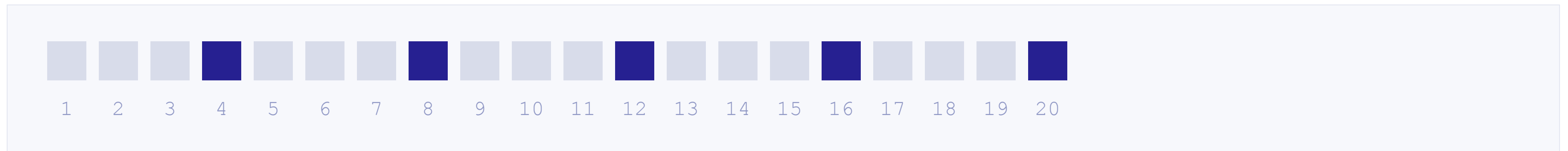
Every unit in the frame has an **equal** chance of selection, and every possible sample of size n is equally likely.

- **How:** number the frame 1...N, then draw n numbers at random without replacement.
- **Good for:** a fairly uniform population and a complete, accessible frame.
- **Weakness:** can miss a small subgroup entirely, purely by luck.

```
df.sample(n=400, random_state=42)
```

TECHNIQUE 02 Systematic sampling

Pick a random start, then take every k -th unit down the list. Here $N = 20$, $n = 5$, so $k = 4$ and the random start is 4.



HOW

$k = N / n$. Pick a random start between 1 and k , then step.

GOOD FOR

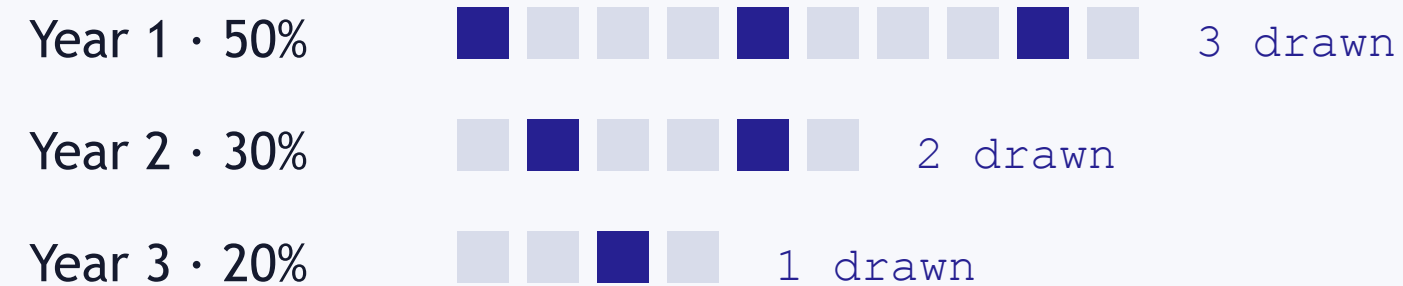
Queues, production lines, long lists — anywhere a full frame is awkward.

WATCH OUT

If the list has a repeating cycle of length k , you sample the same kind of unit every time.

TECHNIQUE 03 Stratified sampling

Split the population into non-overlapping groups (strata), then sample randomly *within each one* , in proportion to its size.



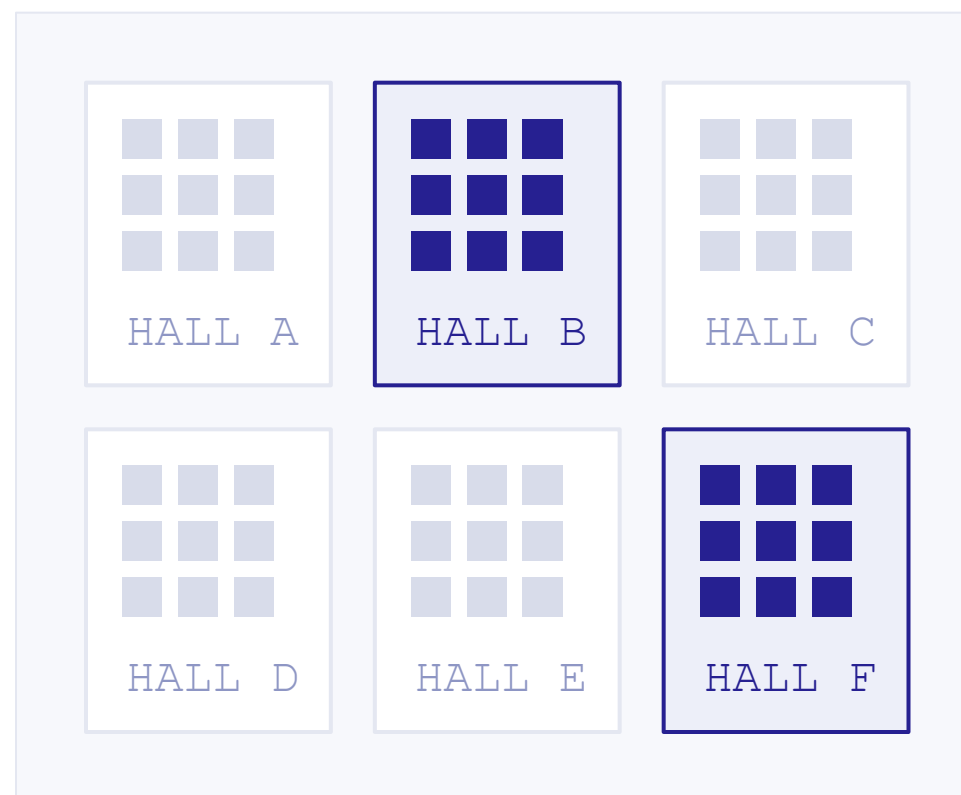
Every stratum is represented, in its true proportion.

- **Good for:** populations with meaningful subgroups – year, gender, campus, income band.
- **Pay-off:** more precise than simple random for the same n, and small groups cannot vanish.
- **Cost:** you must know each unit's stratum before you sample.

```
df.groupby("year", group_keys=False) \
    .apply(
        lambda g: g.sample(frac=0.05))
```

TECHNIQUE 04 Cluster sampling

Divide the population into naturally occurring groups, randomly choose whole groups, and measure everyone inside them.



Stratified samples *some* units from *every* group.
Cluster samples *every* unit from *some* groups.

- **Good for:** geographically spread populations where travel is the main cost.
- **Pay-off:** by far the cheapest probability method – no full frame of individuals needed.
- **Cost:** people in a cluster resemble each other, so precision is lower than simple random at the same n.

Non-probability:

Convenience

Whoever is easiest to reach — the class in front of you, your followers.

Fast and free. Almost always unrepresentative.

Quota

Fill fixed counts per group — 50 men, 50 women — by any means.

Looks like stratified. Isn't — the pick inside each quota is not random.

Purposive

Hand-pick units that are especially informative — experts, extreme cases.

Right choice for depth. Wrong choice for averages.

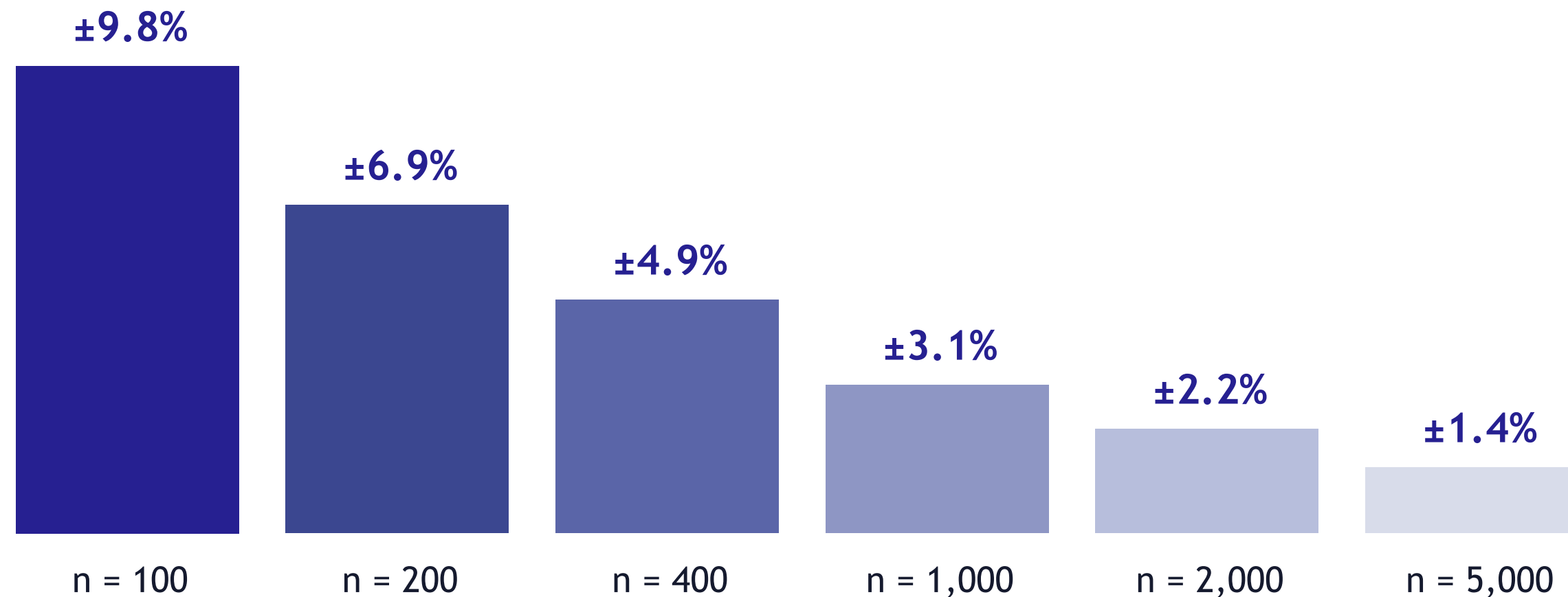
Snowball

Each participant refers the next. Used for hidden or hard-to-reach groups.

Reaches the unreachable — but travels along one social network.

How big does the sample need to be?

Margin of error at 95% confidence, for a large population and the worst case $p = 0.5$.



MARGIN OF ERROR

$$\text{MoE} = 1.96 \times \sqrt{p(1-p) / n}$$

Error falls with the **square root** of n .
Halving it costs four times the sample.

Which is why a national poll of 1,000 works whether the country has 5 million people or 200 million. Population size barely enters the formula.

03

How samples go wrong

Five ways bias creeps in

Undercoverage	Part of the population is missing from the frame entirely.	A phone survey in 1936 — only wealthy households had telephones.
Self-selection	People choose whether to take part, and the willing differ from the rest.	Online reviews: mostly the delighted and the furious.
Non-response	You drew a fair sample, but a specific kind of person never replied.	Night-shift workers never answer a daytime call.
Survivorship	You only see the units that made it far enough to be measured.	Studying successful startups tells you nothing about why others failed.
Measurement	The instrument or the wording pushes answers in one direction.	“Don't you agree the library hours are too short?”

REMEMBER Bias is *systematic* error. More data does not remove it — it just makes you more confident in the wrong answer.

CASE STUDY 2.4 million answers, and still wrong

The 1936 US presidential election. A magazine ran the largest poll ever attempted but a statistician ran a small one.

THE BIG POLL

2.4M

responses returned, from 10 million ballots mailed

Predicted a comfortable win for the challenger. The magazine folded within two years.

THE SMALL POLL

~50K

responses, but drawn to match the electorate

Called the result correctly. Roosevelt won in a landslide, taking 46 of 48 states.

WHAT WENT WRONG

Undercoverage — the mailing list came from car and telephone registries, in a depression year.

Non-response — roughly three in four never replied, and repliers leaned one way.

Sample size cannot rescue a biased frame.

04

Data quality

Six dimensions of data quality

Accuracy

Does the value match reality?

`age = 214`

Completeness

Is anything missing that should be there?

`df.isna().sum()`

Consistency

Do the same facts agree across rows and sources?

`"Lagos" vs "LAGOS"`

Timeliness

Is it recent enough for the decision at hand?

`last updated 2019`

Validity

Does it obey the rules of its own field — type, range, format?

`score = "N/A"`

Uniqueness

Is each real-world unit recorded exactly once?

`df.duplicated().sum()`

Spot the six problems

STUDENT_ID	AGE	CITY	SCORE	ENROLLED
S-001	19	Lagos	72	2024-09-01
S-002	214	Abuja	65	2024-09-01
S-003	21	LAGOS	N/A	2024-09-03
S-004	20	—	81	01/09/2024
S-001	19	Lagos	72	2024-09-01

- ACCURACY age 214 is impossible
- CONSISTENCY Lagos and LAGOS are one city
- VALIDITY "N/A" in a numeric column forces the whole column to text
- COMPLETENESS city is blank for S-004
- VALIDITY two different date formats in one column
- UNIQUENESS S-001 is recorded twice

Why anyone pays for quality

1 Cost to **verify** a record as it is captured.

10 Cost to **clean** it later, in the pipeline.

100 Cost of a **decision** made on it while it is still wrong.

The “1-10-100 rule” – a rule of thumb in data management, not a measured constant.

Analysts routinely report spending the majority of a project on finding and fixing data problems – not on modelling.

Industry surveys put the average annual cost of poor data quality to a single organisation in the millions of dollars. The figures vary wildly by source; the direction never does.

Every hour spent designing collection properly is an hour you do not spend cleaning – and a decision you do not get wrong.

LAB 01 Audit the quality first

CELL 1

```
import pandas as pd

df = pd.read_csv("students.csv")
print(df.shape)           # size
print(df.isna().sum())    # completeness
print(df.duplicated().sum()) # uniqueness
print(df.dtypes)         # validity
```

OUTPUT

```
(500, 5)
student_id    0
age           0
city          12
score         31
dtype: int64
7
```

HOW TO READ IT

- 500 rows, 5 columns, is that the number you expected?
- If out of 500 rows, 31 rows have missing scores, find out *why* before you drop them.
- It's also very important to remove duplicates

HINT

If a numeric column shows dtype `object`, it may mean that some text got mixed up in that column. Find it with `pd.to_numeric(df["score"], errors="coerce").isna()` .

Clean it, then draw the sample

CELL 2 – CLEAN

```
df = df.drop_duplicates()

df["city"] = df["city"].str.strip().str.title()

df["score"] = pd.to_numeric(df["score"],
errors="coerce" )

df = df[df["age"].between(15, 80)]
```

CELL 3 – SAMPLE

```
# simple random
srs = df.sample(n=100, random_state=42)
# systematic, every 5th row
sys = df.iloc[2::5]
# stratified, 10% of each city
strat = df.groupby("city", group_keys=False ) \
          .apply(
lambda g: g.sample(frac=0.1))
```

LAB 03

Prove the sample is fair: Compare entire population with simple random

CELL 4

```
compare = pd.DataFrame({"population":  
df["city"].value_counts(normalize=True),  
"sample": srs["city"].value_counts(normalize=True)  
})  
print(compare.round(3))
```

	population	sample
Lagos	0.412	0.400
Abuja	0.283	0.290
Kano	0.187	0.180
Ibadan	0.118	0.130

WHAT GOOD LOOKS LIKE

- In the previous slide we define srs as random 100 samples from the entire population (df). Comparing their variations tell you how close or far your sample is from reality.
- A gap of ten points or more on a large group means something is wrong with the draw.
- Repeat the comparison for age and score using .mean().

A sample you have not compared against the population is a sample you are simply hoping is fair.

Class work

- 01 Load `50_startups_messy.xlsx` and report its shape, missing counts and duplicate count.
- 02 Clean it: drop duplicates, standardise `State`, coerce `Marketing Spend` to numeric.
- 03 Draw a simple random sample of 20 with `random_state=7`, and a stratified 20% sample by `State`.
- 04 Compare the mean `score` of R&D Spend and Marketing Spend in both samples against the population. Which is closer, and why?

Check in

- Q1 What is the difference between a population and a sampling frame?
-
- Q2 Give one situation where secondary data is the better choice, and one where it is not.
-
- Q3 In one sentence each explain stratified versus cluster sampling.
-
- Q4 Name three of the six quality dimensions and the pandas check for each.